Sequential Conditional Means of Equicorrelated Scores

One, two or three fixed starting scores included in the average

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Each test has population mean 0 and standard deviation 1. Fix the first observed score at a. For jointly Gaussian tests with equal pairwise correlation c, recursively predict each subsequent test by conditioning on the starting score and on all earlier predictions. Every subsequent prediction equals ca. Including a as the first value, the average of $a, e_1, \dots, e_k$ is $a(1 + kc)/(k + 1)$. This fixed recursive average has standard deviation zero. The actual $(k+1)$-test average has population standard deviation $\sqrt{(1 + kc)/(k + 1)}$; after observing the starting score, its conditional standard deviation is $\sqrt{k(1 - c)(1 + kc)}/(k + 1)$. We derive the Gaussian conditional mean, solve the recursion, describe the admissible correlations and singular endpoints, and distinguish these sources of variation. A general derivation also gives the cases with two or three initial scores all fixed at a, including every starting score in the average. Worked examples use initial IQ scores of 150 on a mean-100, SD-15 reporting scale, correlation 0.6, and one, two or ten subsequent predictions. A counterexample shows that means, variances and correlations alone do not determine the recursive conditional means.

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1 Problem and notation

Let $k \geq 1$ be an integer and write

$$X_0, X_1, \dots, X_k$$

for standardized test scores. Throughout this analysis, every test has population mean zero and standard deviation one:

$$\mathrm{E}[X_j] = 0, \qquad \mathrm{SD}(X_j) = 1, \qquad \mathrm{Var}(X_j) = 1, \qquad 0 \leq j \leq k. \tag{1}$$

These parameters describe the reference distribution of each test; a fixed observed score a need not equal zero. We relabel the original tests $A_r(c, 0, 1)$ so that test 0 supplies the starting observation. For distinct indices, assume $\mathrm{Cov}(X_i, X_j) = \mathrm{Corr}(X_i, X_j) = c$.

Use $a \in \mathbb{R}$ for the *fixed starting score*; it is a scalar, not a function. For each $1 \leq n \leq k$, let g_n be the Gaussian conditional mean function of X_n given $(X_0, \dots, X_{n-1})$. Define

$$e_1 = g_1(a) = \mathrm{E}[X_1 \mid X_0 = a], \tag{2}$$

$$\begin{aligned} e_n &= g_n(a, e_1, \dots, e_{n-1}) \\ &= \mathrm{E}[X_n \mid X_0 = a, X_1 = e_1, \dots, X_{n-1} = e_{n-1}], \qquad 2 \leq n \leq k. \end{aligned} \tag{3}$$

The quantity of interest is

$$\boxed{B_k = \frac{a + \sum_{n=1}^{k} e_n}{k+1}.} \tag{4}$$

It contains $k+1$ values: the starting score a is the first value, followed by the k predictions $e_1, \dots, e_k$. Thus k counts subsequent predictions, while $k+1$ counts all values in the average. Section 6 extends this setup to r initial scores all fixed at a, followed by k predictions, for $r + k$ total values.

For continuous distributions, conditioning on a particular score is not a ratio of probabilities of point events. A conditional mean is ordinarily defined only almost everywhere. Here, g_n means the continuous version supplied by Gaussian regression, derived below. This convention specifies the recursion even at individual histories of probability zero. The singular case $c = 1$ will be treated directly on its support.

2 The joint model and its correlation domain

Assume $(X_0, \dots, X_k)$ is jointly Gaussian, allowing degenerate Gaussian distributions at the endpoints. Its covariance matrix is

$$R_{k+1} = (1-c)I_{k+1} + c\mathbf{1}_{k+1}\mathbf{1}_{k+1}^{\mathsf{T}}, \tag{5}$$

where I_d is the d-dimensional identity and $\mathbf{1}_d$ is a column of d ones. Vectors orthogonal to $\mathbf{1}_{k+1}$ have eigenvalue $1-c$; the vector $\mathbf{1}_{k+1}$ has eigenvalue $1+kc$. Thus a covariance matrix for these $k+1$ scores exists exactly when

$$\boxed{-\frac{1}{k} \le c \le 1.} \tag{6}$$

An arbitrarily long equicorrelated collection requires $0 \le c \le 1$. A negative fixed c cannot satisfy the lower bound for all k.

For $0 \le c \le 1$, a construction consistent with the hypothetical common-factor illustration in the repository's Hoeflin/Ferguson analysis, Appendix A, is

$$X_j = \sqrt{c}\, G + \sqrt{1-c}\, \varepsilon_j, \tag{7}$$

where $G, \varepsilon_0, \dots, \varepsilon_k$ are independent standard-normal variables. The covariance-matrix derivation below also covers the admissible negative correlations. This is a mathematical model, not a claim that actual test batteries satisfy these assumptions.

3 Derivation of the conditional mean

Fix $1 \le n \le k$, initially with $c < 1$, and put

$$Y_n = (X_0, \dots, X_{n-1})^{\mathsf{T}}, \qquad R_n = \mathrm{Cov}(Y_n) = (1-c)I_n + c\mathbf{1}_n\mathbf{1}_n^{\mathsf{T}}.$$

The eigenvalue of R_n in the direction $\mathbf{1}_n$ is

$$d_n = 1 + (n-1)c.$$

It is positive throughout $-1/k \le c < 1$: for negative c, $d_n \ge 1 - (n-1)/k \ge 1/k > 0$. The other eigenvalues are $1 - c > 0$. Consequently every conditioning block is nonsingular, even if the full $(k+1)$-score distribution is singular at $c = -1/k$.

An explicit inverse, verified by multiplying it by R_n, is

$$R_n^{-1} = \frac{1}{1-c}\left(I_n - \frac{c}{1+(n-1)c}\mathbf{1}_n\mathbf{1}_n^{\mathsf{T}}\right). \tag{8}$$

Since $\mathrm{Cov}(X_n, Y_n) = c\mathbf{1}_n^{\mathsf{T}}$, the regression coefficient is

$$\mathrm{Cov}(X_n, Y_n)R_n^{-1} = \frac{c}{1+(n-1)c}\mathbf{1}_n^{\mathsf{T}}. \tag{9}$$

To establish why this coefficient gives the conditional mean, define the residual

$$W_n = X_n - \frac{c}{1+(n-1)c}\sum_{j=0}^{n-1} X_j. \tag{10}$$

For each $\ell \in \{0, \dots, n-1\}$,

$$\begin{aligned}\mathrm{Cov}(W_n, X_\ell) &= c - \frac{c}{1+(n-1)c}\sum_{j=0}^{n-1}\mathrm{Cov}(X_j, X_\ell)\\ &= c - \frac{c}{1+(n-1)c}[1+(n-1)c] = 0.\end{aligned}$$

The residual and Y_n are jointly Gaussian because they are linear transformations of a Gaussian vector. Their zero cross-covariance implies independence: the Gaussian characteristic function $\exp(-t^{\mathsf{T}}\Sigma t/2)$ factors across covariance blocks when the cross-covariance block is zero. This argument also allows a zero-variance residual. Since $\mathrm{E}[W_n] = 0$, independence gives $\mathrm{E}[W_n \mid Y_n] = 0$. Taking the conditional expectation in (10) therefore yields

$$\boxed{g_n(y_0, \cdots, y_{n-1}) = \frac{c}{1+(n-1)c}\sum_{j=0}^{n-1} y_j.} \tag{11}$$

This linear function is the unique continuous version, because the nonsingular vector Y_n has a strictly positive density everywhere.

For completeness, the same residual calculation gives

$$\begin{aligned}\mathrm{Var}(X_n \mid Y_n) = \mathrm{Var}(W_n) &= 1 - \frac{nc^2}{1+(n-1)c}\\ &= \frac{(1-c)(1+nc)}{1+(n-1)c}.\end{aligned} \tag{12}$$

This nonnegative conditional variance describes actual test outcomes, not randomness in the recursively specified numbers e_n.

4 Solving the recursion and taking the average

Inserting the prescribed history into (11) gives

$$e_n = \frac{c}{1+(n-1)c}\left(a + \sum_{j=1}^{n-1} e_j\right), \qquad 1 \le n \le k, \tag{13}$$

where the sum is empty at $n = 1$. In particular, $e_1 = ca$. Suppose $e_j = ca$ for all $1 \le j < n$. Then

$$e_n = \frac{c}{1 + (n-1)c}[a + (n-1)ca] = \frac{ca[1 + (n-1)c]}{1 + (n-1)c} = ca.$$

Induction establishes

$$\boxed{e_1 = e_2 = \cdots = e_k = ca.} \tag{14}$$

Consequently the requested average is

$$\boxed{B_k = \frac{a + \sum_{n=1}^{k} e_n}{k+1} = \frac{1 + kc}{k+1}\, a = \left(c + \frac{1-c}{k+1}\right) a.} \tag{15}$$

Unlike the individual subsequent predictions, the requested average depends on the number of values included. With $\lambda_k = (1 + kc)/(k + 1)$, it is $B_k = \lambda_k a$. Once a, c, k are fixed, this average is deterministic. An additional expectation changes nothing: $\mathrm{E}[B_k] = \lambda_k a$. If $N = k+1$ denotes the *total* number of values, the same result is

$$\boxed{B_{N-1} = \frac{1 + (N-1)c}{N}\, a, \qquad N \ge 2.} \tag{16}$$

For the single-value case $N = 1$, the average is simply a, without any subsequent prediction or inter-test correlation being needed.

There is also a partial-sum derivation. Define

$$T_0 = a, \qquad T_n = a + \sum_{j=1}^{n} e_j.$$

Equation (13) implies

$$T_n = \frac{1 + nc}{1 + (n-1)c} T_{n-1}.$$

Multiplying these relations telescopes, without dividing by c or by a:

$$T_n = a \prod_{j=1}^{n} \frac{1 + jc}{1 + (j-1)c} = a(1 + nc).$$

Dividing $T_k = a + \sum_{j=1}^{k} e_j = a(1 + kc)$ by $k + 1$ again gives (15). This proof includes $a = 0$ and $c = 0$.

For example, if $a = 2$ and $c = 0.6$, then

$$e_1 = 1.2, \qquad e_2 = \frac{0.6}{1.6}(2 + 1.2) = 1.2, \qquad e_3 = \frac{0.6}{2.2}(2 + 1.2 + 1.2) = 1.2.$$

Every subsequent prediction remains 1.2. Including the starting score, the average after three predictions is instead

$$B_3 = \frac{2 + 1.2 + 1.2 + 1.2}{4} = \frac{1 + 3(0.6)}{4}\, 2 = 1.4.$$

5 Standard deviation of the mean

5.1 The specified recursive average with fixed starting score

The numbers $a, e_1, \ldots, e_k$ and their average B_k are fixed once a, c, k are specified. Hence

$$\mathrm{Var}(B_k) = \mathrm{E}[(B_k - \mathrm{E}[B_k])^2] = 0, \qquad \boxed{\mathrm{SD}(B_k) = 0.} \tag{17}$$

The actual test-score mean also includes the first test. Its population and conditional standard deviations are derived next.

5.2 Population standard deviation of the actual test-score mean

Define the random mean of all $k+1$ scores by

$$C_k = \frac{1}{k+1}\sum_{j=0}^{k} X_j. \tag{18}$$

Its population mean is zero. There are $k+1$ variance terms, each equal to 1, and $k(k+1)/2$ distinct covariance pairs, each equal to c. Consequently

$$\begin{aligned}\operatorname{Var}(C_k) &= \frac{1}{(k+1)^2}\left(\sum_{j=0}^{k}\operatorname{Var}(X_j) + 2\sum_{0\le i<j\le k}\operatorname{Cov}(X_i, X_j)\right)\\ &= \frac{(k+1)+k(k+1)c}{(k+1)^2} = \frac{1+kc}{k+1} = \lambda_k.\end{aligned} \tag{19}$$

Taking the nonnegative square root gives

$$\boxed{\operatorname{SD}(C_k) = \sqrt{\frac{1+kc}{k+1}} = \sqrt{c + \frac{1-c}{k+1}}.} \tag{20}$$

For $N = k+1$ total tests, this is

$$\boxed{\operatorname{SD}(C_{N-1}) = \sqrt{\frac{1+(N-1)c}{N}}.} \tag{21}$$

The population variance and standard deviation require only the specified finite variances and covariances; joint Gaussianity is not needed for (19)–(21).

5.3 Conditional standard deviation after observing the first score

Return to the joint Gaussian model and fix $X_0 = a$. For $1 \le i \le k$, write $U_i = X_i - cX_0$. The vector of residuals is jointly Gaussian with X_0, has zero covariance with X_0, and is therefore independent of it. Direct calculation gives

$$\begin{aligned}\operatorname{Var}(U_i) &= 1 - c^2,\\ \operatorname{Cov}(U_i, U_j) &= c - c^2 \qquad (i \ne j).\end{aligned}$$

Thus each future score has conditional mean ca, conditional variance $1-c^2$, and each distinct future pair has conditional covariance $c - c^2$. Including the fixed first score gives

$$\operatorname{E}[C_k \mid X_0 = a] = \frac{a + kca}{k+1} = \lambda_k a = B_k. \tag{22}$$

The starting score contributes no variance after it is fixed, so

$$\begin{aligned}\operatorname{Var}(C_k \mid X_0 = a) &= \frac{k(1-c^2) + k(k-1)(c-c^2)}{(k+1)^2}\\ &= \frac{k(1-c)(1+kc)}{(k+1)^2} = \lambda_k(1-\lambda_k).\end{aligned} \tag{23}$$

Therefore

$$\boxed{\mathrm{SD}(C_k \mid X_0 = a) = \frac{\sqrt{k(1-c)(1+kc)}}{k+1}.} \tag{24}$$

This conditional standard deviation does not depend on the numerical value of a under the Gaussian model. The value of a determines the conditional mean. In terms of the total count $N = k + 1$,

$$\mathrm{SD}(C_{N-1} \mid X_0 = a) = \frac{\sqrt{(N-1)(1-c)[1+(N-1)c]}}{N}. \tag{25}$$

For the example $a = 2, c = 0.6, k = 3$, the four-score mean satisfies

$$\mathrm{E}[C_3 \mid X_0 = 2] = 1.4, \qquad \mathrm{SD}(C_3) = \sqrt{0.7} \approx 0.836660,$$

$$\mathrm{SD}(C_3 \mid X_0 = 2) = \sqrt{0.21} \approx 0.458258, \qquad \mathrm{SD}(B_3) = 0.$$

The population spread, the uncertainty remaining after the first score is observed, and the variation of the fixed recursive mean are different quantities.

5.4 Predictors based on actual earlier outcomes

The random predictors

$$P_n = \mathrm{E}[X_n \mid X_0, \dots, X_{n-1}]$$

use actual prior outcomes and are not the deterministic values e_n. The tower property implies $\mathrm{E}[P_n \mid X_0] = cX_0$, while its unconditional implication $\mathrm{E}[P_n] = 0$ holds without Gaussianity whenever the scores are integrable and have mean zero. Applying a conditional mean function at a fixed, recursively chosen history is a different operation.

5.5 The subsequent predictions considered separately

For comparison, averaging just the k subsequent predictions gives

$$\frac{1}{k}\sum_{n=1}^{k} e_n = ca. \tag{26}$$

The requested average B_k includes a as well and therefore has denominator $k + 1$, as in (15).

5.6 Population mean zero and the fixed starting score

If one randomizes the starting observation according to its standardized test distribution, the recursively filled average becomes $\lambda_k X_0$. Consequently

$$\mathrm{E}[\lambda_k X_0] = 0, \qquad \mathrm{SD}(\lambda_k X_0) = \lambda_k, \tag{27}$$

where $\lambda_k \geq 0$ throughout the admissible correlation domain. These are population statements obtained by varying the starting score. In particular, the standard deviation λ_k of the recursively filled mean

across random starting scores generally differs from the standard deviation $\sqrt{\lambda_k}$ of the actual test-score mean. The law of total variance accounts for the difference:

$$\begin{aligned}\operatorname{Var}(C_k) &= \operatorname{Var}(\operatorname{E}[C_k \mid X_0]) + \operatorname{E}[\operatorname{Var}(C_k \mid X_0)] \\ &= \lambda_k^2 + \lambda_k(1-\lambda_k) = \lambda_k.\end{aligned}$$

For the fixed starting score in this problem, the answer is $B_k = \lambda_k a$, which need not be zero. Every original test still has mean 0 and standard deviation 1; the arithmetic average has its own distribution and is not rescaled to unit standard deviation.

6 Two or three initial scores equal to the starting value

6.1 General setup with several fixed starting scores

Let $r \geq 1$ be the number of starting scores and keep $k \geq 1$ for the number of subsequent predictions. There are now $N = r + k$ jointly Gaussian scores $X_0, \ldots, X_{N-1}$, each with population mean 0, standard deviation 1, and pairwise correlation c. Fix the first r scores at the same value:

$$X_0 = X_1 = \cdots = X_{r-1} = a. \tag{28}$$

These are r distinct tests with equal observed scores. Define

$$d_r = 1 + (r-1)c, \qquad d_N = 1 + (N-1)c.$$

The covariance-matrix eigenvalues give the domain

$$-\frac{1}{N-1} \leq c \leq 1. \tag{29}$$

For $c < 1$, all conditioning prefixes are nonsingular. In particular,

$$d_r \geq 1 - \frac{r-1}{N-1} = \frac{k}{N-1} > 0.$$

As before, evaluation at an exact history means the continuous Gaussian conditional-mean version. This specifies the equal-score histories even though their point probabilities are zero.

6.2 Every subsequent prediction and the inclusive average

Let $e_j^{(r)}$ be the recursive prediction of X_{r+j-1}, using the r fixed initial scores and the previous $j-1$ predictions. Equation (11), applied to the $r + j - 1$ preceding scores, gives

$$e_j^{(r)} = \frac{c}{d_r + (j-1)c}\left(ra + \sum_{\ell=1}^{j-1} e_\ell^{(r)}\right), \qquad 1 \leq j \leq k. \tag{30}$$

The first prediction is $b_r = rca/d_r$. If the previous predictions all equal b_r, then

$$ra + \sum_{\ell=1}^{j-1} e_\ell^{(r)} = ra + (j-1)\frac{rca}{d_r} = \frac{ra}{d_r}[d_r + (j-1)c].$$

Substituting into (30) and cancelling the positive denominator establishes the induction step. Consequently

$$\boxed{e_j^{(r)} = b_r = \frac{rc}{1+(r-1)c}\,a, \qquad 1 \le j \le k.} \tag{31}$$

The requested average includes all r starting values and all k subsequent predictions, so

$$\begin{aligned} B_{r,k} &= \frac{ra + \sum_{j=1}^{k} e_j^{(r)}}{r+k} = \frac{ra + k(rca/d_r)}{N} \\ &= \boxed{\frac{r\,[1+(r+k-1)c]}{(r+k)[1+(r-1)c]}\,a.} \end{aligned} \tag{32}$$

Putting $r = 1$ recovers (15). Holding all starting scores, c, and the counts fixed makes $B_{r,k}$ deterministic, with $\mathrm{SD}(B_{r,k}) = 0$.

6.3 Population and conditional standard deviations

Let the actual mean of the $N = r + k$ test outcomes be

$$C_{r,k} = \frac{1}{N}\sum_{i=0}^{N-1} X_i.$$

The covariance count in (19) gives

$$\boxed{\mathrm{SD}(C_{r,k}) = \sqrt{\frac{1+(r+k-1)c}{r+k}} = \sqrt{\frac{d_N}{N}}.} \tag{33}$$

This population formula does not require Gaussianity. The following conditional calculation does use the joint Gaussian model.

Let $Z = (X_0, \dots, X_{r-1})^{\mathsf{T}}$. Since $\mathrm{Cov}(Z) = (1-c)I_r + c\mathbf{1}_r\mathbf{1}_r^{\mathsf{T}}$, the projection of each future score onto the starting block is $(c/d_r)\sum_{i=0}^{r-1} X_i$. Define the future residuals

$$V_j = X_{r+j-1} - \frac{c}{d_r}\sum_{i=0}^{r-1} X_i, \qquad 1 \le j \le k.$$

Each residual has zero covariance with every coordinate of Z. Joint Gaussianity makes the residual vector independent of Z. Its covariance entries are

$$\begin{aligned} \mathrm{Var}(V_j) &= 1 - \frac{rc^2}{d_r}, \\ \mathrm{Cov}(V_i, V_j) &= c - \frac{rc^2}{d_r} = \frac{c(1-c)}{d_r} \qquad (i \ne j). \end{aligned}$$

Equivalently, the conditional covariance matrix of the future scores is

$$(1-c)I_k + \frac{c(1-c)}{d_r}\mathbf{1}_k\mathbf{1}_k^{\mathsf{T}}.$$

Conditioning on $Z = a\mathbf{1}_r$ fixes all r starting terms. The future conditional means are all rca/d_r, so $\mathrm{E}[C_{r,k} \mid Z = a\mathbf{1}_r] = B_{r,k}$. Summing the residual covariance terms gives

$$\begin{aligned}\mathrm{Var}(C_{r,k} \mid Z = a\mathbf{1}_r) &= \frac{1}{N^2}\left[k(1-c) + \frac{k^2c(1-c)}{d_r}\right] \\ &= \frac{k(1-c)(d_r+kc)}{N^2 d_r} = \frac{k(1-c)d_N}{N^2 d_r}.\end{aligned} \tag{34}$$

Thus the conditional standard deviation of the actual mean is

$$\boxed{\mathrm{SD}(C_{r,k} \mid Z = a\mathbf{1}_r) = \frac{1}{r+k}\sqrt{\frac{k(1-c)[1+(r+k-1)c]}{1+(r-1)c}}.} \tag{35}$$

It is independent of the numerical value of a. This is the spread of the actual mean after the starting block is observed; the fixed recursive mean itself has standard deviation zero. An across-population calculation with random initial scores uses a random starting *block*. The single random-start interpretation in (27) does not specify a distribution for several starting scores constrained to share one random value.

6.4 Special case: two starting scores equal to a

Set $r = 2$. The initial values are a, a, followed by k predictions, so $N = k + 2$ and $-1/(k+1) \le c \le 1$. Every prediction and the inclusive average are

$$e_j^{(2)} = \boxed{\frac{2ca}{1+c}}, \qquad 1 \le j \le k, \tag{36}$$

$$B_{2,k} = \frac{2a + \sum_{j=1}^{k} e_j^{(2)}}{k+2} = \boxed{\frac{2[1+(k+1)c]}{(k+2)(1+c)}\,a.} \tag{37}$$

The actual mean has population and conditional standard deviations

$$\mathrm{SD}(C_{2,k}) = \boxed{\sqrt{\frac{1+(k+1)c}{k+2}}}, \tag{38}$$

$$\mathrm{SD}(C_{2,k} \mid X_0 = X_1 = a) = \boxed{\frac{1}{k+2}\sqrt{\frac{k(1-c)[1+(k+1)c]}{1+c}}.} \tag{39}$$

6.5 Special case: three starting scores equal to a

Set $r = 3$. The initial values are a, a, a, followed by k predictions, so $N = k + 3$ and $-1/(k+2) \le c \le 1$. The predictions and inclusive average are

$$e_j^{(3)} = \boxed{\frac{3ca}{1+2c}}, \qquad 1 \le j \le k, \tag{40}$$

$$B_{3,k} = \frac{3a + \sum_{j=1}^{k} e_j^{(3)}}{k+3} = \boxed{\frac{3[1+(k+2)c]}{(k+3)(1+2c)}\,a.} \tag{41}$$

The actual mean has standard deviations

$$\mathrm{SD}(C_{3,k}) = \boxed{\sqrt{\frac{1+(k+2)c}{k+3}}}, \tag{42}$$

$$\mathrm{SD}(C_{3,k} \mid X_0 = X_1 = X_2 = a) = \boxed{\frac{1}{k+3}\sqrt{\frac{k(1-c)[1+(k+2)c]}{1+2c}}}. \tag{43}$$

For both special cases, the standard deviation of the fixed recursive average $B_{r,k}$ is 0.

6.6 Checks, endpoints and an example

At $c = 0$, every subsequent prediction is zero, $B_{r,k} = ra/(r+k)$, and the actual mean has population SD $1/\sqrt{r+k}$ and conditional SD $\sqrt{k}/(r+k)$. At $c = 1$, all test scores agree almost surely. The equal starting history is compatible with this support; all predictions and their inclusive mean equal a. The population SD is 1, and the conditional SD is 0, establishing the endpoint directly.

At $c = -1/(N-1)$, the total score is zero almost surely. The initial block remains nonsingular because $k \geq 1$, and

$$e_j^{(r)} = -\frac{ra}{k}, \qquad ra + \sum_{j=1}^{k} e_j^{(r)} = 0.$$

Hence $B_{r,k} = 0$, and both actual-mean standard deviations are zero. If there are no subsequent predictions, the fixed list of r values a simply averages to a, with conditional SD zero, provided the starting history is compatible with the model. The general derivation above assumes $k \geq 1$; it does not divide by a singular initial-block eigenvalue when $k = 0$. For $r \geq 2$ and $k = 0$, the lower endpoint $c = -1/(r-1)$ permits equal starting scores only when $a = 0$.

For $a = 2$, $c = 0.6$, and $k = 3$ subsequent predictions, the two special cases give the following values, rounded to six decimal places:

Quantity	Two initial scores	Three initial scores
Each subsequent prediction	1.500000	1.636364
Mean including all starting scores	1.700000	1.818182
Population SD of the actual mean	0.824621	0.816497
Conditional SD of the actual mean	0.319374	0.246183
SD of the fixed recursive mean	0	0

7 Worked examples starting at IQ 150

7.1 Convert the reporting scale to the standardized model

The model scores X_i continue to have population mean 0 and standard deviation 1. To report them on an IQ scale with mean 100 and standard deviation 15, define

$$Q_i = 100 + 15X_i. \tag{44}$$

An initial observed IQ of 150 therefore corresponds to

$$a = \frac{150-100}{15} = \frac{10}{3}. \tag{45}$$

This conversion preserves the tests' correlations. The mean and SD of the standardized model are unchanged; Q_i is its reporting scale. In the examples below, $r \in \{1, 2, 3\}$ is the number of initial IQ-150 scores, $k \in \{1, 2, 10\}$ is the number of subsequent predictions, and every initial score is included in the $N = r + k$ values being averaged. Use the selected correlation $c = 0.6 = 3/5$ throughout this section.

7.2 Predictions, means and standard deviations in IQ points

Put $d_r = 1 + (r-1)c$ and $d_N = 1 + (N-1)c$, as above. Converting (31) gives the IQ prediction for every subsequent score:

$$q_r = 100 + 15\frac{rca}{d_r} = 100 + 50\frac{rc}{d_r}. \tag{46}$$

At $c = 0.6$, these predictions are

$$q_1 = 130, \qquad q_2 = 137.5, \qquad q_3 = \frac{1550}{11} \approx 140.909091. \tag{47}$$

They depend on the number of fixed starting scores r, and are constant across all k subsequent recursive predictions within a case.

The recursive average in IQ points, also equal to the conditional expected value of the actual IQ-score mean, is

$$m^{\mathrm{IQ}}_{r,k} = 100 + 15B_{r,k} = 100 + 50\frac{rd_N}{Nd_r} = \frac{150r + kq_r}{r + k}. \tag{48}$$

Adding 100 changes a mean, but does not change a standard deviation. Multiplication by 15 multiplies both means relative to 100 and standard deviations by 15. Hence the actual IQ-score mean has population and conditional standard deviations

$$\sigma^{\mathrm{IQ}}_{\mathrm{pop},r,k} = 15\sqrt{\frac{d_N}{N}}, \tag{49}$$

$$\sigma^{\mathrm{IQ}}_{\mathrm{cond},r,k} = \frac{15}{N}\sqrt{\frac{k(1-c)d_N}{d_r}}. \tag{50}$$

The conditional SD is calculated after all r initial IQ-150 scores have been observed. The fixed recursively filled average itself has SD 0.

7.3 The nine requested examples

The following table reports the inclusive mean and both actual-mean SDs in IQ points, rounded to two decimal places. Here $N = r + k$ includes every starting score and every subsequent prediction. The mean is $m^{\mathrm{IQ}}_{r,k}$ from (48); the SD columns use (49) and (50).

Initial scores r	Subsequent predictions k	Total N	Mean IQ	Population SD	Conditional SD
1	1	2	140.00	13.42	6.00
1	2	3	136.67	12.85	6.63
1	10	11	131.82	11.97	7.22
2	1	3	145.83	12.85	3.71
2	2	4	143.75	12.55	4.44
2	10	12	139.58	11.94	5.45
3	1	4	147.73	12.55	2.68
3	2	5	146.36	12.37	3.34
3	10	13	143.01	11.91	4.46

For example, with two starting scores of 150 and ten subsequent predictions, each subsequent IQ prediction is 137.5. The inclusive mean is therefore

$$\frac{2(150) + 10(137.5)}{12} = \frac{1675}{12} \approx 139.583333.$$

In this row, the population variance of the actual IQ mean is 285/2, and its conditional variance is 475/16. Their square roots give approximately 11.937336 and 5.448624 IQ points, respectively. The fixed recursive mean in the same row remains a deterministic number.

8 Boundary cases with one starting score

At $c = 0$, joint Gaussianity makes the scores independent, and every subsequent prediction is zero. Including the starting score gives $B_k = a/(k+1)$. The population and conditional standard deviations of the actual mean are then $1/\sqrt{k+1}$ and $\sqrt{k}/(k+1)$, respectively. At $c = 1$, $\mathrm{Var}(X_i - X_j) = 2(1-c) = 0$, so all scores agree almost surely. Given the starting score a, every compatible recursive prediction is $a = ca$, and $B_k = a$. This establishes the result at $c = 1$ without using the singular inverse in (8). The population standard deviation of C_k is 1, while its conditional standard deviation after observing $X_0 = a$ is 0.

At the other endpoint, $c = -1/k$, the full vector satisfies

$$\sum_{j=0}^{k} X_j = 0 \quad \text{almost surely,}$$

because that sum has mean zero and variance $(k+1)(1+kc) = 0$. Every conditioning prefix remains nonsingular, as shown above, so the derivation of (15) applies. Its prescribed history is compatible with the support, since $a + kca = 0$, so $B_k = 0$. The final conditional variance (12) is zero. Both the population and conditional standard deviations of C_k are also zero at this endpoint. The fixed recursive mean has standard deviation zero at every admissible correlation. For the single-value case $N = 1$, the actual score has population standard deviation 1, and fixing its value leaves conditional standard deviation 0.

9 Why correlation alone is insufficient

The original description specifies only marginal means, marginal variances and pairwise correlations. Those quantities do not determine the conditional mean functions. A counterexample already exists with one subsequent test. Let

$$P(X_0 = -\sqrt{2}) = P(X_0 = \sqrt{2}) = \tfrac{1}{4}, \qquad P(X_0 = 0) = \tfrac{1}{2}, \qquad X_1 = X_0^2 - 1.$$

Symmetry and direct calculation give

$$\mathrm{E}[X_0] = 0, \quad \mathrm{E}[X_0^2] = 1, \quad \mathrm{E}[X_0^3] = 0, \quad \mathrm{E}[X_0^4] = 2.$$

Consequently

$$\begin{aligned} \mathrm{E}[X_1] &= \mathrm{E}[X_0^2] - 1 = 0, \\ \mathrm{Var}(X_1) &= \mathrm{E}[(X_0^2 - 1)^2] = 2 - 2 + 1 = 1, \\ \mathrm{Cov}(X_0, X_1) &= \mathrm{E}[X_0^3 - X_0] = 0. \end{aligned}$$

Both scores are standardized and $c = 0$, yet at the fixed starting value $a = 0$, an event with positive probability,

$$e_1 = \mathrm{E}[X_1 \mid X_0 = 0] = -1 \neq 0 = ca.$$

The average including the starting value is therefore

$$B_1 = \frac{0 + (-1)}{2} = -\frac{1}{2},$$

whereas (15) would give zero under the Gaussian model. Therefore (15) is a result under the stated joint Gaussian model, or another model with the same conditional-mean functions; it does not follow from equicorrelation alone.

Scope and provenance. This derivation and its AI review extend the repository's hypothetical Gaussian model. Existing Ada/SPARK and Python/Deal evidence concerns the specified software; it does not formally verify this text. No new empirical study or formal-prover campaign was performed. The accompanying README records build and visual checks. The repository's original material follows its existing license, subject to third-party rights.